Applications of Taylor expansion

- For practice we denote $h = x - x_0$
- Find the Taylor expansion of $sin(x)$ about $x_0 = \pi/2$

Answer:

Derivative d(sin(x)) = cos(x) and d(cos(x)) = -sin(x)

$$\sin(x) = \sin(\pi/2) + h\cos(\pi/2) - \frac{h^2}{2!}\sin(\pi/2) - \frac{h^3}{3!}\cos(\pi/2)$$

$$+\frac{h^4}{4!}\sin(\pi/2) + \frac{h^5}{5!}\cos(\pi/2) - \ldots$$

where $h = x - \pi/2$

Derivative d(tan(x)) = sec(2x) and d(sec(2x) = 2.sec^2 (x).tan (x)

Find the Taylor expansion of *tan*(*x*) about x0=π/2

Applications of Taylor expansion

- With the help of previous page;
- Find the Taylor expansion of $cos(x)$ about $x_0 = \pi/3$

Hints: $h = x - x_0$, $d(sin(x)) = cos(x)$ and $d(cos(x)) = -sin(x)$

Answer:

Calculating Truncation Error

Example 1:

Find the Taylor expansion of e^x about $x_0 = 0$ which use the first two, three, four and five and evaluate for $x = 0.5$, respectively.

Answer: Given, $x_0 = 0$, $\therefore h = (x - x_0) = x$

$$e^x = 1 + x + \mathrm{O}(x^2)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \mathrm{O}(x^3)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \mathrm{O}(x^4)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \mathrm{O}(x^5)$$

Calculating Truncation Error

$$e^{0.5} = 1.648721..$$ (True value)

$$e^{0.5} = 1.5 + O(x^2) \Rightarrow O(x^2) \cong 0.14872..$$

$$e^{0.5} = 1.625 + O(x^3) \Rightarrow O(x^3) \cong 0.023721..$$

$$e^{0.5} = 1.64583.. + O(x^4) \Rightarrow O(x^4) \cong 0.002887..$$

$$e^{0.5} = 1.64843.. + O(x^5) \Rightarrow O(x^5) \cong 0.0002837..$$

Calculating Truncation Error

Example 2:

Find the Taylor expansion of e^x about $x_0 = 0.25$ which use the first two, three, four and five and evaluate for $x = 0.5$, respectively.

Answer: Given, $x_0 = 0.25$, $\therefore h = (x - x_0) = (0.5 - 0.25) = 0.25$

$$e^x = e^{0.25} + he^{0.25} + O(h^2) \qquad \text{since } (d(e^x))/(dx) = e^x$$

$$e^x = e^{0.25} + he^{0.25} + \frac{h^2}{2!}e^{0.25} + O(h^3)$$

$$e^x = e^{0.25} + he^{0.25} + \frac{h^2}{2!}e^{0.25} + \frac{h^3}{3!}e^{0.25} + O(h^4)$$

$$e^x = e^{0.25} + he^{0.25} + \frac{h^2}{2!}e^{0.25} + \frac{h^3}{3!}e^{0.25} + \frac{h^4}{4!}e^{0.25} + O(h^5)$$

Calculating Truncation Error

$$e^{0.5} = 1.648721..$$

$$e^{0.5} = 1.60503.. + O(h^2) \Rightarrow O(h^2) \cong 0.04368..$$

$$e^{0.5} = 1.64515.. + O(h^3) \Rightarrow O(h^3) \cong 0.00356..$$

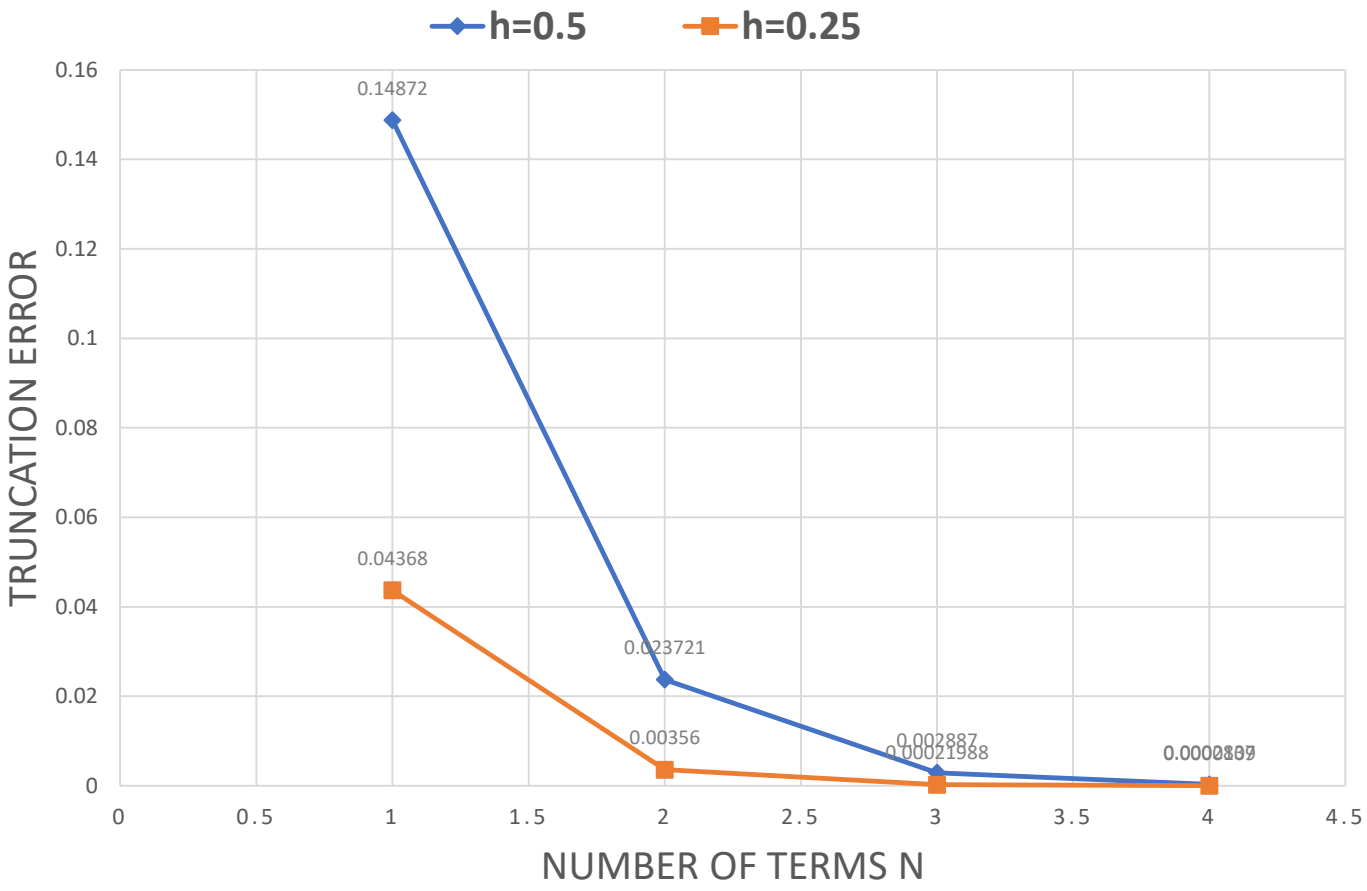
$$e^{0.5} = 1.64850.. + O(h^4) \Rightarrow O(h^4) \cong 2.1988..e-004$$

$$e^{0.5} = 1.64871.. + O(h^5) \Rightarrow O(h^5) \cong 1.0900..e-005$$

Summary from example 1 and 2:

Order of Trunc. Er.	$x_0=0,\ h=0.5$	$x_0=.25,\ h=0.25$	*Ratio*
$O(h^2)$	*0.14872..*	*0.04368..*	*3.4*
$O(h^3)$	*0.023721..*	*0.00356..*	*6.66*
$O(h^4)$	*0.002887..*	$2.1988..x10^{-4}$	*13.13*
$O(h^5)$	*0.0002837..*	$1.090..x10^{-5}$	*26.03*

Truncation error *vs n*



More Example

Use Taylor series expansions with $n = 0$ to 6 to approximate $f(x) = cos(x)$ at $x_{i+1} = \pi/3$ on the basis of the value of $f(x)$ and its derivatives at $x_i = \pi/4$. Note that this means that $h = \pi/3 - \pi/4 = \pi/12$.

Ans.: $f(\pi/3) = cos(\pi/3) = 0.5$ (True value)

- Considering result on 1st value only: $f(\pi/3) \cong \cos(\pi/4) = 0.707106781$

  which represents a percent relative error of

  $\varepsilon_t = ((|0.5 - 0.707106781|)/0.5)\times 100\% = 41.4213562\%$

- Taking 1st 2 values: $f(\pi/3) \cong \cos(\pi/4) - \pi/12\times \sin(\pi/4)$

  $= 0.707106781 - 0.2617993878 \times 0.707106781 = 0.521986658625$

  $\varepsilon_t = ((|0.5 - 0.521986658625|)/0.5) \times 100\% = 4.397331725\%$

Example …

- Taking 1st 3 values:

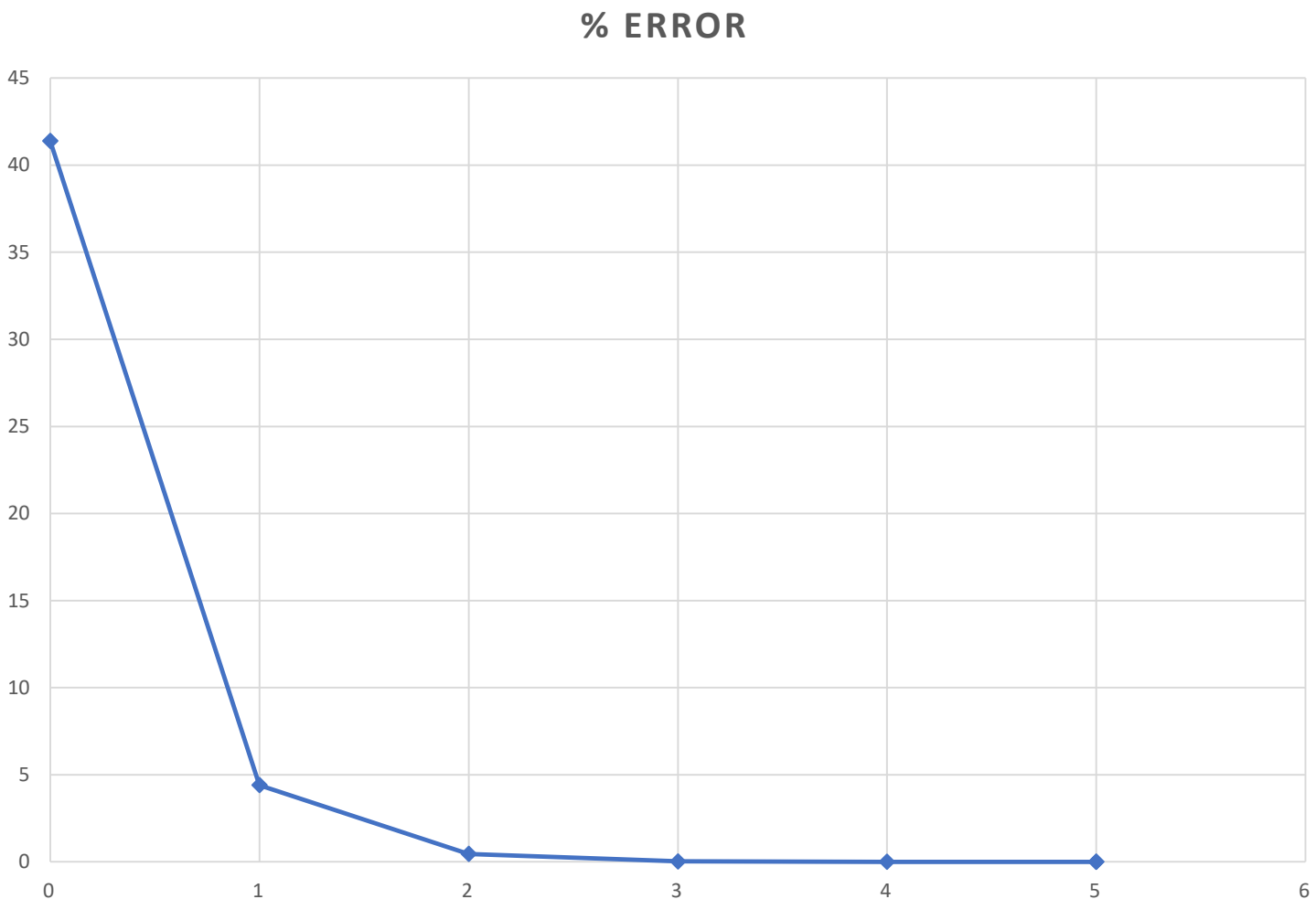
$$f(\pi/3) \cong \cos(\pi/4) - \pi/12 \times \sin(\pi/4) - (\pi/12)^2/2 \times \cos(\pi/4)$$

$$= 0.707106781 - 0.2617993878 \times 0.707106781$$

$$- (0.2617993878)^2 \times 0.707106781/2 = 0.4977544913$$

$$\varepsilon_t = ((|0.5 - 0.4977544913|)/0.5) \times 100\% = 0.4491017457\%$$

Example …

n	$f(\pi/3)=0.5(True)$	% Error
0	0.707106781	41.4213562
1	0.521986659	4.397331725
2	0.497754491	0.4491017457
3	0.499869147	2.62×10^{-2}
4	0.500007551	1.51×10^{-3}
5	0.500000304	6.08×10^{-5}
6	0.499999988	2.44×10^{-6}

Example …